Chapter No. 1

INTRODUCTION

The presence of a skull fracture following head trauma can lead to several complications, and increase the risk of an underlying subdural or epidural hematoma (de Boussard et al., 2006). Therefore, the detection of skull fractures is very important for the evaluation of traumatic head injury. Nowadays, the cranial computed tomography (CT) has become the diagnostic standard of care for suspected skull or brain injury (for Acute Care , UK et al.(2007). The radiologist can diagnose the presence of skull fractures based on CT. All these features will makethe manual detection time-consuming and laborious. Thus, presenting an efficient automatic skull fracture detecting method is very significant. Some methods have been already proposed for the skull fracture detecting. H.Shao et al. (Shao and Zhao, 2003) proposed region growing method to segment the brain and then used the entropy function to generate the rules for the skull fracture diagnosis. Wan et al. (Zaki et al., 2009) applied Sobel edge detection technique for the identification of skull fracture. Ayum et al. (Yamada et al., 2018) proposed a surface selective Black-Hat transform method for the skull fracture detection, and reached a sensitivity of 86%. However, all the methods mentioned above only considered the local features of the fractures, and there were many false positives among the prediction results.Recently, deep learning models, especially the convolutional neural networks (CNNs) have been widely applied into the image processing. Different types of skull fractures are as shown in fig 1.1. Compare to traditional methods, CNNs can extract the high-dimensional features of the image and have shown excellent performance in the object detection, image classification and segmentation tasks (Ronneberger et al., 2015; Ren et al., 2015; Liu et al., 2016; He et al., 2017). In the area of object detection, the CNN-based methods can be divided into two kinds: the region-proposal networks, like Faster R-CNN (Ren et al., 2015); the region-free networks, like SSD (Liu et al., 2016) and YOLOV3 (Redmon and Farhadi, 2018). Compared to the latter kind, Faster R-CNN takes more time on detecting, but obtains more accurate results especially for the small objects. Although Faster R-CNN have been proved that it could well detect the objects in natural images, but there are some problems when it is applied into the skull fracture detecting directly. First, lots of the ground truth boxes of the fractures have very small size, thus, suitable candidate regions are difficult to be obtained. Moreover, the time consumption may be very large, because for one CT scan, there are more than a hundred of 2D slices. Aiming at the skull fracture detection, we propose a novel CNN-based model, named as Skull R-CNN. Guiding by the morphological features of the skull, a skeleton-based region proposal method

is designed for more reasonable candidate regions. On the other hand, a novel full resolution feature network is constructed, which can optimize the feature extraction for the small objects. The experiments results show that SkullR-CNN achieve the AP of 0.65 on the validation set, 0.60 on the test set and reduce time consumption by 60% than Faster R-CNN. Compared the previous works on the skull fracture detection, Skull R-CNN significantly reduces the less false positives, and keep a high sensitivity.MRI scan of the skull is shown in the fig 1.2.

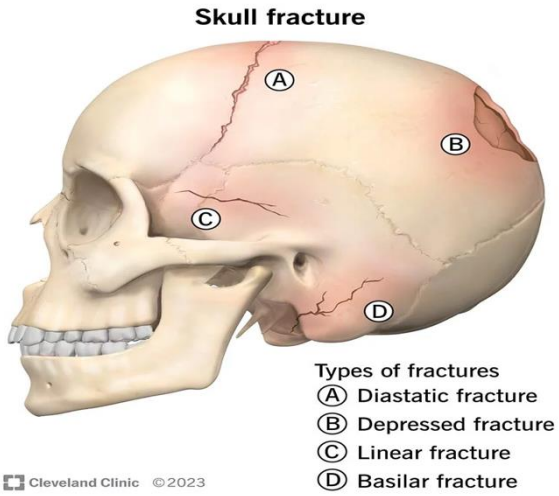

Figure 1.1 Different types of skull fractures

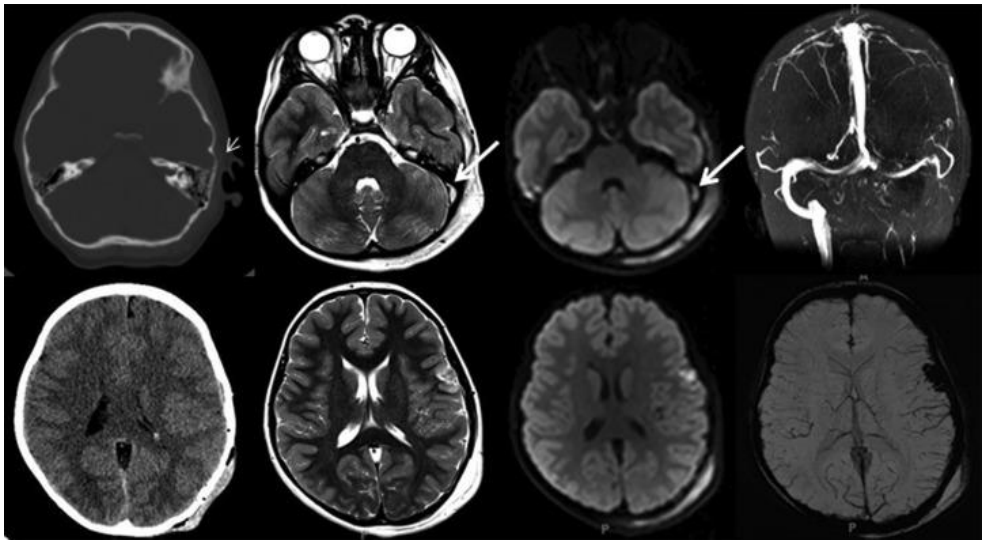

Figure 1.2 MRI Scan of a skull fracture

Chapter No. 2

LITERATURE SURVEY

Automated Identification of Skull Fractures With Deep Learning et.al [1] This study addresses the life-threatening complications of skull fractures by proposing a deep learning system for automated identification from cranial CT scans. Analyzing 4,782 patient scans (development set: 4,168, test set: 614), including data from 7,856 healthy individuals, the models were trained using two approaches: object detection with YOLOv3 and segmentation with a modified attention U-net. Manual labeling by seven neurologists validated the results. On a test set of 235 patients, YOLOv3 achieved 80.64% sensitivity and 85.92% specificity, while the modified attention U-Net reached 82.80% sensitivity and 88.73% specificity. The study concludes that deep learning effectively identifies skull fractures, favoring the segmentation approach for potential improved results.

In a March 2022 research, Sozan Mahammed Ahmed et.al [2] this retrospective multicenter study, conducted in two tertiary hospitals in Seoul, aimed to assess the diagnostic value of brain computed tomography (CT) alone in detecting facial fractures and its role in determining the need for additional facial CT. Analyzing data from 860 adult patients with blunt trauma to the head and face, who underwent both brain and facial CT on the same day, the study found a statistically significant predominance of facial fractures in men without a specific age group bias. The sensitivity, specificity, positive predictive value, negative predictive value, and accuracy of brain CT for detecting upper facial bone fractures were 83.72%, 98.87%, 97.17%, 92.92%, and 94.08%, respectively. These results suggest that brain CT holds high diagnostic value and accuracy, aiding emergency physicians in deciding whether additional facial CT is necessary.

This Post-mortem computed tomography (PMCT) et.al [3] systematic review aimed to conduct a meta-analysis of the sensitivity and specificity of post-mortem computed tomography (PMCT) in detecting skull fractures, considering the increasing use of PMCT in forensic pathology. The review included 18 eligible studies published between January 2000 and August 2021, involving 1538 cases. Excluding studies on children, sharp trauma, gunshot injuries, natural deaths, case reports, and reviews, the meta-analysis indicated a PMCT sensitivity of 0.87 [0.80; 0.92] for detecting skull base fractures, with a specificity of 0.96 [0.90; 0.98]. For vault fractures, the sensitivity was 0.89 [0.80; 0.94], and specificity was 0.96 [0.91; 0.98]. Limitations include the variability in samples across studies. Overall, the findings

suggest that PMCT exhibits good sensitivity and specificity in detecting skull fractures in post-mortem examinations.

In this study, the authors Yangling Ma, Yixin Luo et.al [4] propose a comprehensive system for bone fracture detection in X-ray images, addressing the challenge of identifying fractures across various anatomical regions. The method employs Faster R-CNN for initial bone detection, utilizing bounding boxes and classification to identify different types of bones. Subsequently, a Crack-Sensitive Convolutional Neural Network (Crack Net) is introduced to specifically detect fractures within different types of bones. This two-stage approach aims to alleviate the burden on doctors by providing possible fracture regions in X-ray images. The system offers bone segmentation and fracture region detection, as illustrated in Figure 1. Notably, the previous methods focused on single anatomical regions or specific fracture types, whereas this proposed system demonstrates universal capabilities across diverse bone types. Additional results and experiments are detailed in the supplementary section.

In this study, the authors Md MoniruzzamanEmonet.al [5] emphasize the critical need for swift identification and classification of brain injuries resulting from skull fractures, highlighting the potential for significant bleeding over major blood vessels. The paper stresses that patients with skull fractures exhibit more intracranial hematomas, necessitating a prompt and accurate diagnosis. Computed Tomography (CT) emerges as the primary diagnostic tool for such injuries, providing detailed information to physicians. However, the manual diagnosis and grading of skull fractures on CT scans prove time-consuming and challenging due to various fracture characteristics. The authors propose an automated skull fracture categorization and detection system to expedite the diagnostic process. This system not only aids in identifying critical patients but also addresses understaffing issues in hospitals, particularly evident in the CQ-500 dataset where radiologists struggled to reach a consensus on fracture cases. The paper underscores the urgency for efficient computational approaches to process and analyze vast amounts of CT scan data for improved diagnostic judgments and treatment planning.

In this study, the authorsKorean J Radiolet.al [6] developed and assessed a deep learning-based artificial intelligence (AI) model for detecting pediatric skull fractures on plain radiographs. The retrospective multi-center study included a development dataset from two hospitals and an external test set from a third hospital. The reference standard for skull fractures was cranial computed tomography (CT). The AI model exhibited an impressive area under the receiver operating characteristic curve (AUROC) of 0.922 in the internal test set and 0.870 in the external test set. Sensitivity and specificity for the internal test set were 81.1% and 91.3%,

respectively, and for the external test set were 78.9% and 88.2%, respectively. Notably, the AI model significantly improved diagnostic performance for inexperienced radiologists and emergency physicians but showed no significant impact on a pediatric radiologist. The findings highlight the potential of AI in enhancing diagnostic accuracy for pediatric skull fractures on plain radiographs.

In this prospective study, the authors Niccolo` Parr et.al [7] aimed to assess the accuracy of bedside emergency ultrasound in diagnosing skull fractures in pediatric patients with head trauma, comparing it to computed tomography (CT) scan results. A consecutive series of children requiring CT scan for head trauma were enrolled, and cranial bedside ultrasound imaging was performed by an emergency physician. The study found that bedside ultrasound exhibited 100% sensitivity (95% CI 88.2–100%) and 95% specificity (95% CI 75.0–99.9%) when compared to CT scan for diagnosing skull fractures. The positive and negative predictive values were 97.2% (95% CI 84.6–99.9%) and 100% (95% CI 80.2–100%), respectively. The authors conclude that bedside ultrasound, with its simplicity, minimal required experience, and absence of ionizing radiation, can be a valuable diagnostic tool for pediatric skull fractures, suggesting its incorporation into minor head injury prediction rules and advocating further investigation.

Sujeet More et.al [8] proposed study that was completed on December 7, 2020 analyzes the usage of machine learning methods. The 150 knee photos in the dataset used for this study are primarily TI Weighted images and are from a public collection. Notably, 74% of the data show that the investigation specifically focused on the former b tibia segments. The method used is complete, integrating modern methods like convolutional neuronal networks (CNN) and deep learning techniques with traditional methods like support vector algorithms and statistical pattern knowledge as a result Remarkably, the accuracy of 89% was attained. With this accomplishment, the report admits several shortcomings. The sensitivity to changes in imaging circumstances and the conspicuous lack of comparison studies with other cutting-edge techniques present difficulties.

The traditional diagnostic process for skull fractures et.al [9] involves time-consuming manual analysis of CT scans and radiology reports by a radiologist, followed by treatment decisions made by physicians. As a severe head impact can result in both skull fractures and brain injuries, quick identification of potential brain lesions is crucial for proper treatment. CT scans have become the standard diagnostic method for suspected skull or brain injuries, providing detailed information about brain structures and tissue. However, the characteristics of skull fractures on CT scans, such as appearing as narrow slits, being found in various locations and

lengths, and the presence of tiny fractures, make manual detection and classification challenging. To address this, the study proposes SkullNetV1, a novel deep learning model for automated detection and classification of five common skull fractures, offering potential benefits for efficient medical image retrieval and diagnosis in understaffed hospitals.

In this study, the authorsEva Deininger-Czermak, Andre Euler, Sabine Franckenberg et.al [10] compared the visualization of skull bone and fracture detection between CT and MRI, considering MRI as an alternative modality due to its optimal soft tissue contrast. Twenty subjects underwent CT and MRI within a 72-hour interval, with evaluation conducted on 2D ultrashort echo time (UTE) and 3D multi-echo in-phase fast-field-echo (FRACTURE) sequences. Independent raters assessed qualitative characteristics and fracture detectability in various skull subregions. The interrater and intermodality agreement was measured using intraclass coefficients (ICC). Results indicated good to excellent agreement for both MRI sequences in different subregions, with the highest fracture detection in the skull vault. The study suggests that these MRI sequences may serve as alternatives for surgical planning or follow-up exams for the osseous neurocranium, while acknowledging their inferior performance in visualizing skull base and viscerocranium compared to standard CT imaging.

In a March 2022 research, Sozan Mahammed Ahmed et.al [11] investigated the Bone Segmentation technique as a complete method for treating osteoarthritis in the knee. Using a dataset based on 3D DEK Sagital water excitation of the OAI database and 80 individuals for model construction, the study examined how to create traditional deep learning. With the use of a hybrid Sum-DRF model architecture and Segnet CNN for deep learning and segmentation, the used approach produced an astounding 95.45% accuracy. But the study has several weaknesses that could cause this challenging to evaluate the recommended methods, that is the lack of a quantitative assessment against standard data sets.

Hafsa Khalid et.al [12] examines 32 publications in the literature review on deep learning methods for knee cartilage segmentation in medical pictures; five of the studies particularly address CT scans, while six address MRI using deep learning methods. Three strategies are used in the paper's methodology to automatically recognize and distinguish knee cartilage structures utilizing CNN-based techniques. Metrics including precision, recall, and F1-score are used in the evaluation of the suggested approach to gauge the segmentation algorithm's efficacy and correctness. Deep learning for use in an X-ray (87.7%), MRI examinations (91.82%), and CT imaging (77.6%) has highly accurate stages according to the data. The lack of a thorough examination of computing efficiency and resource requirements—important factors for real-world applications presents a weakness in the work, though.

The research on knee osteoarthritis carried out by E Losina et.al [13] on January 25, 2013, experts in the areas of medical economics and medication economics created economic frameworks from a number of data sources like data from clinical trials, health care claims, and observational research. The method included carefully determining important patient groups with knee osteoarthritis, using medication modifying medical care and control measures, and collecting, examining, and analyzing clinical and financial data. A disadvantage of the methodology is the use of assumptions in their value models, which increases confusion. In these difficulties, the study's 76.35% accuracy rate shows how valuable the research might be for the pharmaceutical economics sectors.

In this study, the authors et.al [14] aimed to evaluate the diagnostic performance of a deep learning algorithm for identifying skull fractures in plain radiographic images and assess its clinical applicability. A total of 2,026 plain radiographic images (fracture, 991; normal, 1,035) from 741 patients were analyzed using the RetinaNet architecture as a deep learning model. Precision, recall, and average precision metrics were employed to measure the algorithm's diagnostic performance. Results indicated varying average precision for different intersection over union (IOU) thresholds, with notable performance differences between anterior-posterior, Towne, and both lateral views. False positives were associated with vascular grooves and suture lines, while false negatives were observed in diastatic fractures, fractures crossing suture lines, and fractures around vascular grooves and the orbit. The study concludes that the deep learning-based object detection algorithm holds promise as a valuable tool in diagnosing skull fractures.

The interest in radiologic efficacyet.al [15] in the context of head trauma began with Bell and Loop's 1971 study, outlining high-yield criteria that successfully identified skull fractures in a significant number of patients. Despite subsequent reports confirming the limited clinical usefulness of skull radiography in trauma settings, the standard practice often includes skull series for most head trauma patients. Phillips refined high-yield criteria, emphasizing that occasional negative skull series can create a false sense of security and pose liability risks. Despite efforts to modify physician behaviour, factors like mentor pressure, patient demands, and time constraints contribute to the continued overuse of skull radiography. At HarborviewMedical Centre, despite access to advanced CT scanners, nearly 2,500 skull radiographs are obtained annually

Chapter No. 3

PROBLEM DEFNITION AND OBJECTIVES

Problem definition

The proposed solution aims to develop a novel method, called Skull R-CNN, based on the Faster R-CNN algorithm, to achieve more accurate and robust skull fracture detection results. The proposed method incorporates a skeleton-based region proposal method guided by the morphological features of the skull to improve the candidate box generation process, and a novel full resolution feature network to obtain more precise features for better detection of small objects like skull fractures.

3.1 Drawbacks of Existing work

Classification of skull fracture is a challenging task for both radiologists and researchers. Skull fractures result in broken pieces of bone, which can cut into the brain and cause bleeding and other injury types. So it is vital to detect and classify the fracture very early. In real world, often fractures occur at multiple sites.

This makes it harder to detect the fracture type where many fracture types might summarize a skull fracture. Unfortunately, manual detection of skull fracture and the classification process is time-consuming, threatening a patient's life. Because of the emergence of deep learning, this process could be automated. Convolutional Neural Networks (CNNs) are the most widely used deep learning models for image categorization because they deliver high accuracy and outstanding outcomes compared to other models. We propose a new model called SkullNetV1 comprising a novel CNN by taking advantage of CNN for feature extraction and lazy learning approach which acts as a classifier for classification of skull fractures from brain CT images to classify five fracture types. Our suggested model achieved a subset accuracy of 88%, an F1 score of 93%, the Area Under the Curve (AUC) of 0.89 to 0.98, a Hamming score of 92% and a Hamming loss of 0.04 for this seven-class multi-labeled classification.

3.2 Objectives

Objectives for the detection of a skull fracture include:

- Accuracy Improvement: Achieve a high level of accuracy in detecting skull fractures from medical imaging data (e.g., X-rays, CT scans) to ensure reliable clinical decision-making.
- Multi-modality Integration: Incorporate multiple types of medical imaging data (e.g., X-rays, CT scans, MRI) to improve detection capabilities and provide a comprehensive assessment of fractures.
- Provide a comprehensive 3D view of the skull, aiding in the understanding of complex fracture patterns.
- Develop new methods and materials for the treatment of skull fractures based on insights gained from 3D models.

CHAPTER No. 4

PROPOSED METHODOLOGY

Figure with Explanation

4.1 Skull R-CNN:

R-CNN, or Region-based Convolutional Neural Network is a general framework for object detection, it can be applied to medical imaging tasks, including the detection of abnormalities like skull fractures. The proposed method integrates the skull morphological features with deep convolutional networks to accurately detect the skull fractures. Below fig 6.1 shows the pipeline of our proposed Skull R-CNN. First, multi-scale feature maps and all candidate region boxes are obtained, then with the help of the Roi-Align (He et al., 2017), the corresponding feature maps of each box is adjusted to the same dimension. Next, the classification module will give a score to each candidate box and screen out suitable boxes. Finally, the position of the boxes will be fined tune to better contain the fractures.

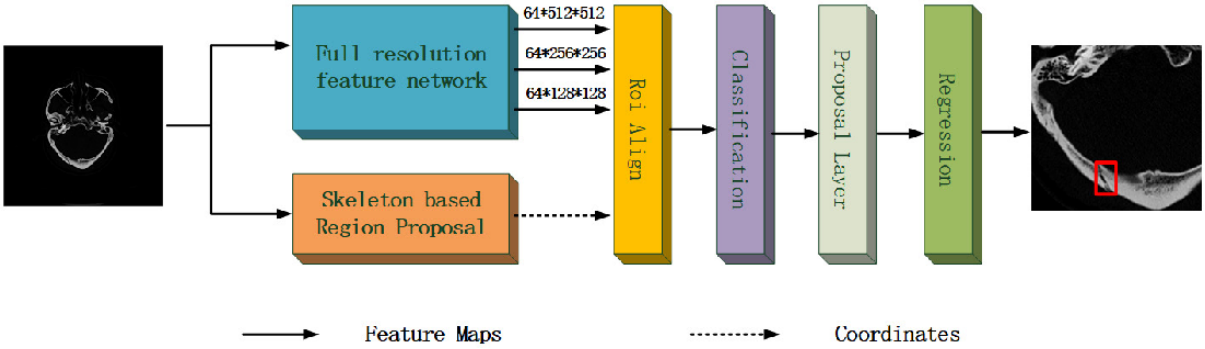

Figure4.1 The architecture of the Skull R-CNN.

In the medical imaging domain, including skull fracture detection, R-CNN can be applied to analyse CT scans or X-ray images. The training process involves providing the algorithm with labelled medical images, indicating the presence or absence of skull fractures. Once trained, the model can then be used to automatically detect and localize skull fractures in new, unseen images. It's important to note that there are more advanced versions of R-CNN, such as Fast R-CNN and Faster R-CNN, which improve the efficiency and speed of the detection process. Additionally, newer architectures like Mask R-CNN can provide not only object detection but also segmentation, which is useful for delineating the exact boundaries of detected fractures.

4.2 Pre-processing:

In the pre-processing phase for skull fracture detection in medical images, several essential steps are undertaken to enhance the quality and relevance of the data. Initially, images are resized to a standardized scale to ensure consistency, followed by normalizing pixel values for improved model training. Contrast workflow of the study shown in fig 6.2.1 enhances techniques, such as histogram equalization, are applied to highlight subtle details, and noise reduction methods help mitigate interference caused by various factors during image acquisition. Windowing is employed to adjust intensity levels and focus on relevant bone structures, while image cropping isolates the specific region of interest, such as the skull. Data augmentation is introduced to diversify the training dataset through transformations like rotation and scaling. For three-dimensional images, anisotropic resampling maintains consistent voxel dimensions. Lastly, accurate labelling of images is crucial, indicating the presence or absence of skull fractures, to train the detection model effectively. These pre-processing steps collectively contribute to refining the input data and optimizing the subsequent skull fracture detection process.

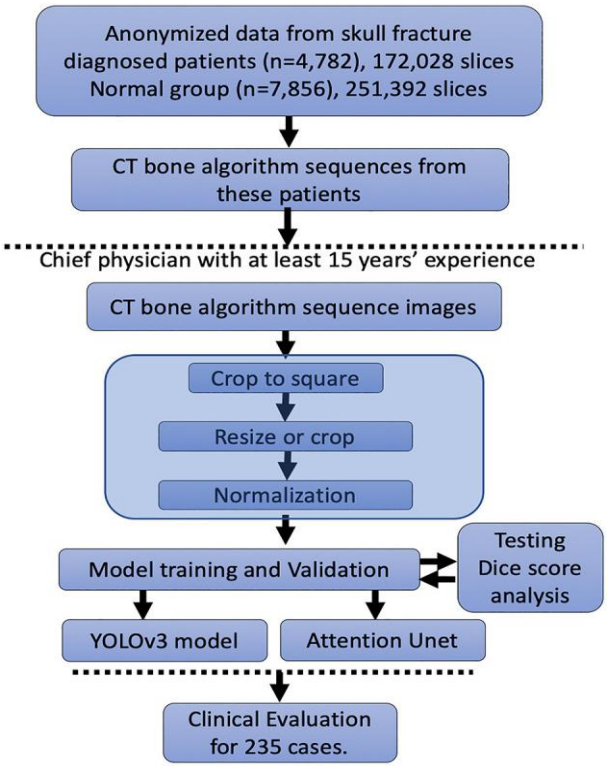

Figure 4.2.1 Workflow of the study

4.3 Feature Extraction:

- For each region proposal, a CNN is used to extract features. In the case of skull fracture detection, the CNN analyses the pixel values within the proposed bounding box to identify relevant patterns and features.
- The features extracted from each region proposal are then used for object classification. The algorithm determines whether the region contains a skull fracture or not.
- This step involves training the system on a dataset with labelled examples of images containing skull fractures and background images without fractures.
- Convolutional layers: The initial layers of the CNN perform convolution operations, where filters are applied to local regions of the input image. These filters help detect basic features such as edges, corners, and textures.
- Deeper convolutional layers capture more complex and abstract features relevant to skull fractures. The network learns to recognize hierarchical patterns, combining low-level features into higher-level representations.

  The network learns to extract features automatically through the training process, where it adjusts the weights of its neurons based on the provided labeled data. The learned features are expected to be representative of the characteristics associated with skull fractures, allowing the model to generalize well to new, unseen images during testing.

4.4 Algorithm details

The YOLOv8architecture was trained with the composite YOLOv8loss function with a batch size of 8 using the Adam optimizer. Image augmentation strategies including resize, crop, zoom, horizontal flip, and rotation were randomly applied during the training process to improve the model generalization performance. The initial learning rate was set to 10∧{-3} and the learning rate was reduced by a factor of 2 if the loss on the validation set (20% of the training data) did not decrease for 10 consecutive epochs. The training was stopped when the validation set loss did not decrease for 50 consecutive epochs and the model with the lowest validation set loss was selected as a final skull fracture object detection model. This model was used to detect the skull fractures on the internal and external test datasets.

The performance of the YOLOv8 architecture was evaluated using precision and recall in correctly identifying the fracture bounding boxes. The box-wise precision was defined as the number of correctly detected fracture bounding boxes divided by the total number of predicted bounding boxes. Similarly, the box-wise recall was defined as the number of correctly detected fracture bounding boxes divided by the total number of ground truth fracture bounding boxes. The precision and recall were separately calculated for every patient and were averaged across all the patients to compute the dataset level performance.

CHAPTER No. 5

IMPLEMENTATION DETAILS

Detection Implementation:

Step 1: Setup and Imports

Import necessary libraries for YOLO (YOLO model), image processing (OpenCV), data manipulation (Pandas, JSON), plotting (Matplotlib, Seaborn), and random sampling.

Set Seaborn plotting style to 'darkgrid'.

Use %matplotlib inline to display plots inline in Jupyter notebooks.

Step 2: Display Random Images with Bounding Boxes

Define paths to training and validation image directories.

List all images in the training directory and select 16 random images.

Create a 4x4 subplot grid using Matplotlib.

For each selected image:

Read the image using OpenCV.

Load the annotations from the JSON file.

Loop through the annotations and draw bounding boxes on the image.

Display the annotated images in the subplot grid.

Step 3: Prepare Training Data Annotations

Initialize lists for bounding boxes (bounds) and images (images).

Define paths for training images and labels.

Load annotations from the JSON file.

For each image in the training set:

Loop through annotations and calculate normalized bounding box coordinates.

Append bounding box data to bounds.

Append image file names to images.

For each image:

Write bounding box data to corresponding label files in YOLO format.

Debugging: Print the contents of a sample label file and the bounding box list.

Step 4: Prepare Validation Data Annotations

Similar steps to training data, but for validation data:

List validation images and initialize bounding box and image lists.

Load validation annotations.

For each image, read and save the image, calculate normalized bounding box coordinates, and append to lists.

Write bounding box data to validation label files.

Print the contents of a sample validation label file.

Step 5: Display an Example Image with Bounding Boxes

Define the path to a sample training image.

Read and display the image using Open

3D Reconstruction

Segmenting and Rendering a 3D Model

Use SimpleITK to read a series of DICOM images.

Perform thresholding segmentation to extract bone structures.

Convert the segmented slices to a NumPy array and use marching cubes to extract surfaces.

Plot the 3D model of segmented bones using matplotlib.

Reading DICOM Files and Extracting Instances

Use glob to read all DICOM files from the train directory.

Extract the InstanceNumber from each DICOM file and store the file paths in the correct order in the instance list.

Creating an Animated GIF from DICOM Images

Define the Img to GIF class to create an animated GIF from the DICOM images.

Adding Images to GIF:

Read each DICOM file, extract the image data, and add it to the GIF.

Creating a 3D Tensor from DICOM Images

Initialize a 3D numpy array to store all the images.

Loop through the instance list, read each DICOM file, and store the pixel data in the 3D array.

Normalization and Visualization:

Define functions for normalizing the data and plotting histograms.

Normalize the array and visualize it using histograms.

Resizing and Plotting 3D Data

Resize the normalized 3D array to a smaller dimension for easier handling.

Define functions for exploding the data (for 3D plotting) and plotting the 3D data.

Advanced 3D Visualization and Manipulation

Create a copy of the resized array and modify it to zero out certain regions for different visualizations.

Segmenting and Rendering a 3D Model

Use Simple ITK to read a series of DICOM images.

Perform thresholding segmentation to extract bone structures.

Convert the segmented slices to a NumPy array and use marching cubes to extract surfaces.

Plot the 3D model of segmented bones using matplotlib.

CHAPTER No. 6

RESULTS AND DISCUSSION

6.1 Datasets

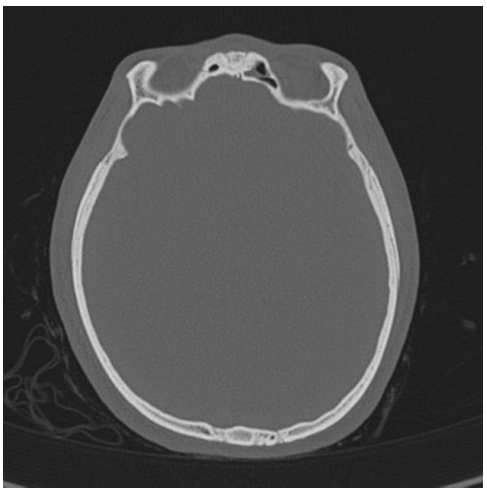

Figure 6.1.1 Normal skull

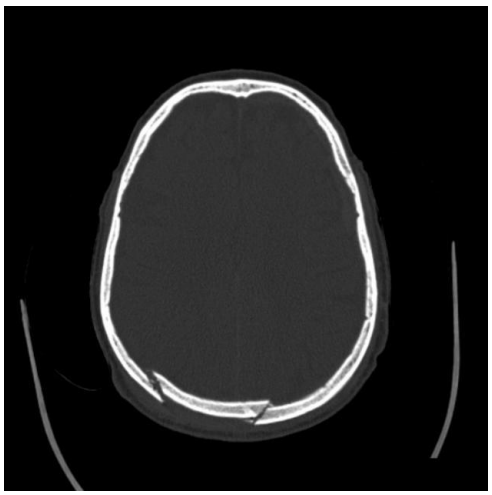

Figure 6.1.2 Fractured skull

6.2 Performance measures

Performance measures are crucial in evaluating the effectiveness and reliability of the skull fracture detection. In the proposed system leveraging Region based Convolution Neural Networks (R-CNNs), several performance measures are considered to ensure the system's accuracy, efficiency, and overall effectiveness.

1. Accuracy: Accuracy is a fundamental performance measure that assesses the model's ability to correctly classify instances into their respective classes (fracture or normal). It represents the percentage of correctly classified instances out of the total instances in the test set. Achieving high accuracy indicates that the model can reliably differentiate between fracture and normal skull images.

2. Confusion Matrix: A confusion matrix provides a detailed breakdown of the model's predictions, including true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). The confusion matrix visually represents the model's performance in terms of correctly and incorrectly classified instances. It is instrumental in understanding the model's strengths and weaknesses, identifying areas for improvement, and fine-tuning the classification thresholds.

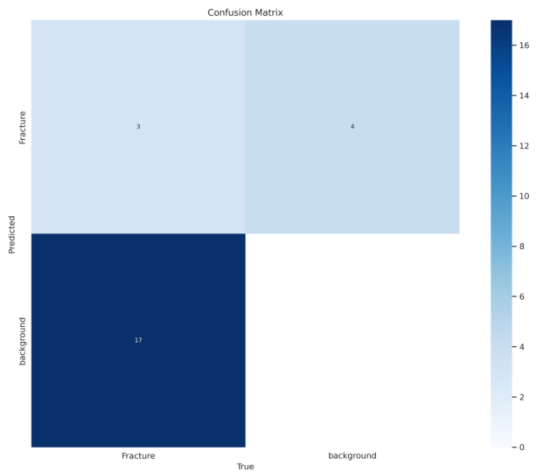

Figure 6.2.1 Confusion Matrix

6.3 Graphs with explanation

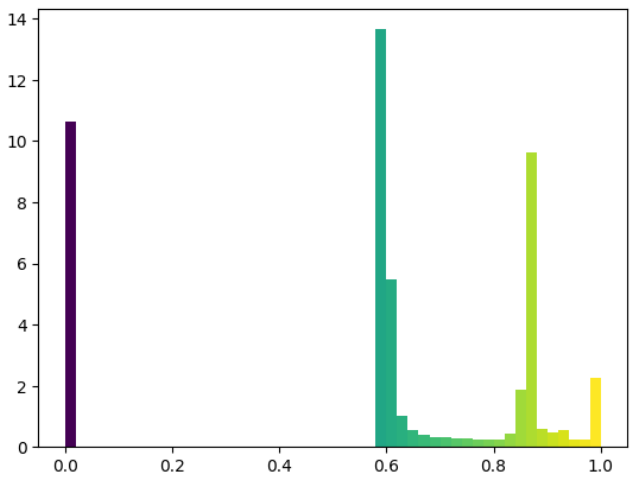

Figure 6.3.1 3D Skull colour Histogram

The above figure shows a 3D histogram would have three axes representing the red, green, and blue intensities of the colors in the image. Each axis would have a range of values from 0 to 255. The x-axis would represent the red intensity, the y-axis would represent the green intensity, and the z-axis would represent the blue intensity. The height of each bar in the 3D histogram would represent the number of pixels in the image that have that particular combination of red, green, and blue intensities.

The 2D bar graph you sent likely shows the distribution of colors in a single image along a grayscale range. The x-axis represents the intensity of the grayscale value, with 0 being black and 1 being white. The y-axis represents the number of pixels in the image that have that particular grayscale intensity.

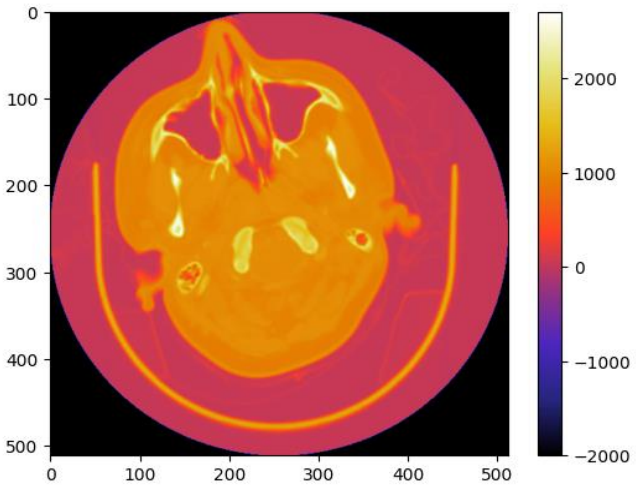

Figure 6.3.2 3D Reconstruction of Fracture detected Skull

The figure 6.3.2 shows different colours that might represent the skull bone, soft tissues, and the fracture line itself and the 3D reconstruction of fracture in slices from different angles.

6.4 Screen shots of results with explanation

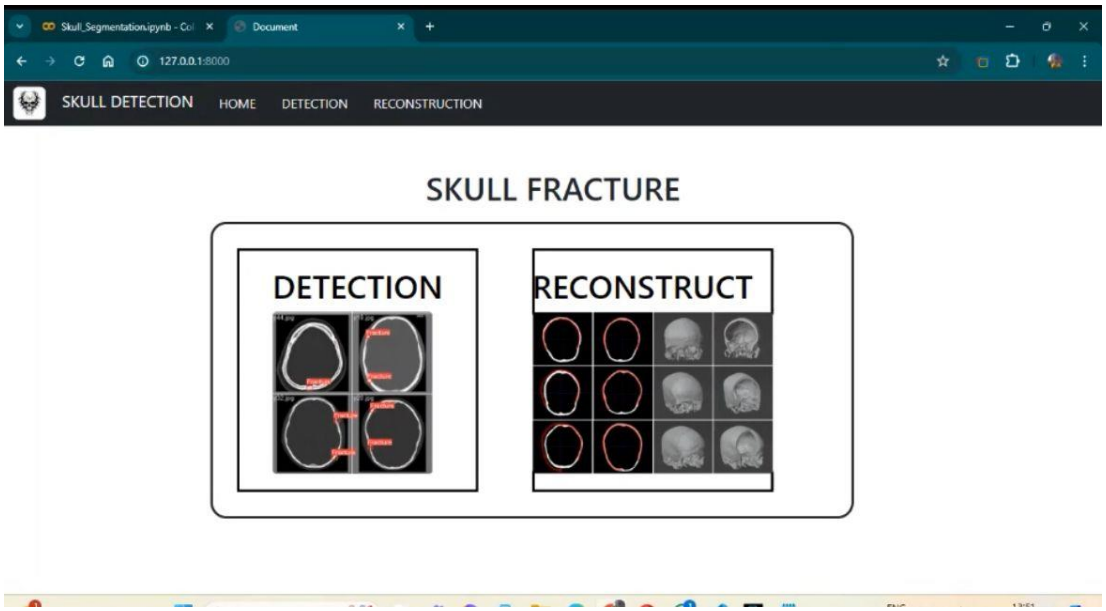

Figure 6.4.1 Application FrontEnd with two Sections, a)Detection and b) Reconstruction

The fig 6.4.1 shows the first page of the application. It contains 2 sections a) Detection : which redirects to the detection part B) 3D reconstruction: which redirects to the reconstruction part.

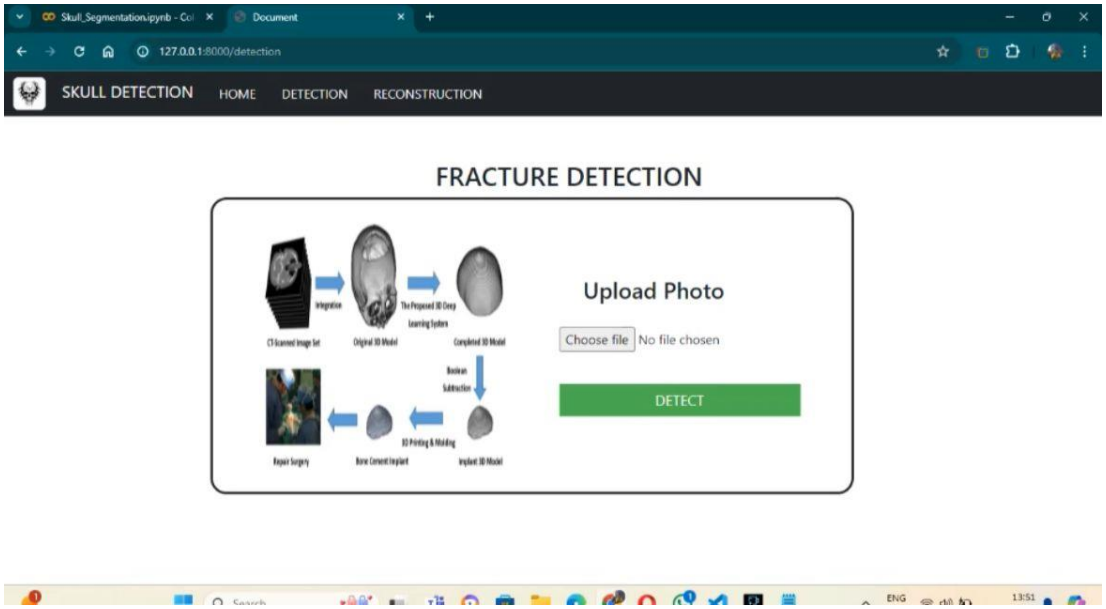

Figure 6.4.2 Fracture Detection Section in Application

The fig 6.4.2 shows the detection page of the application, which has a choose file button, where the skull CT scan image has to be inserted.

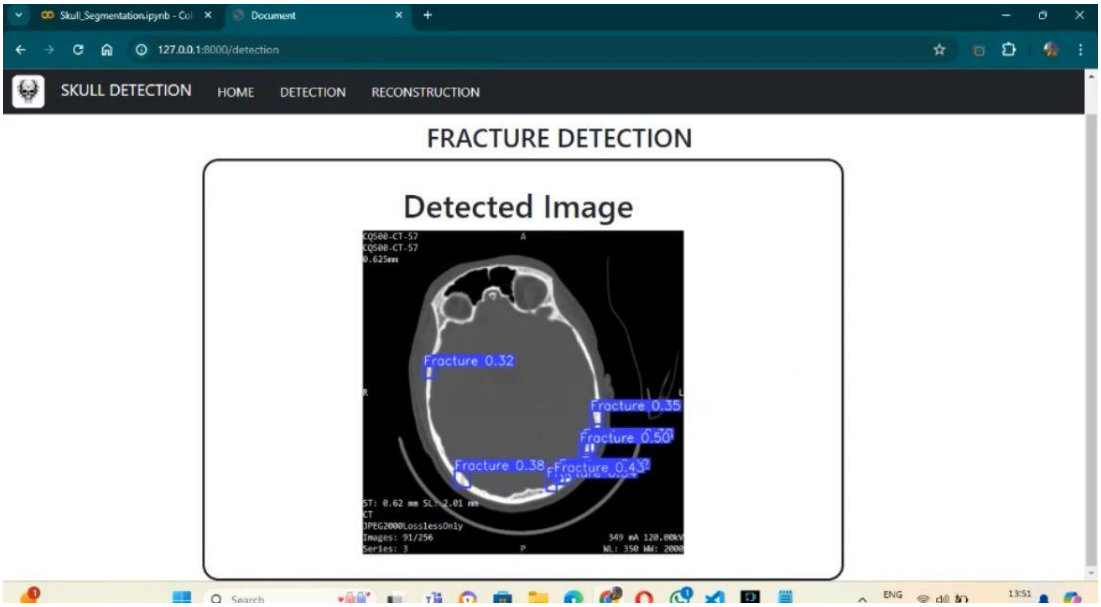

Figure 6.4.3 Fracture Detected Image with mm of the fractures

The figure 6.4.3 shows the output image of the detected skull fracture.

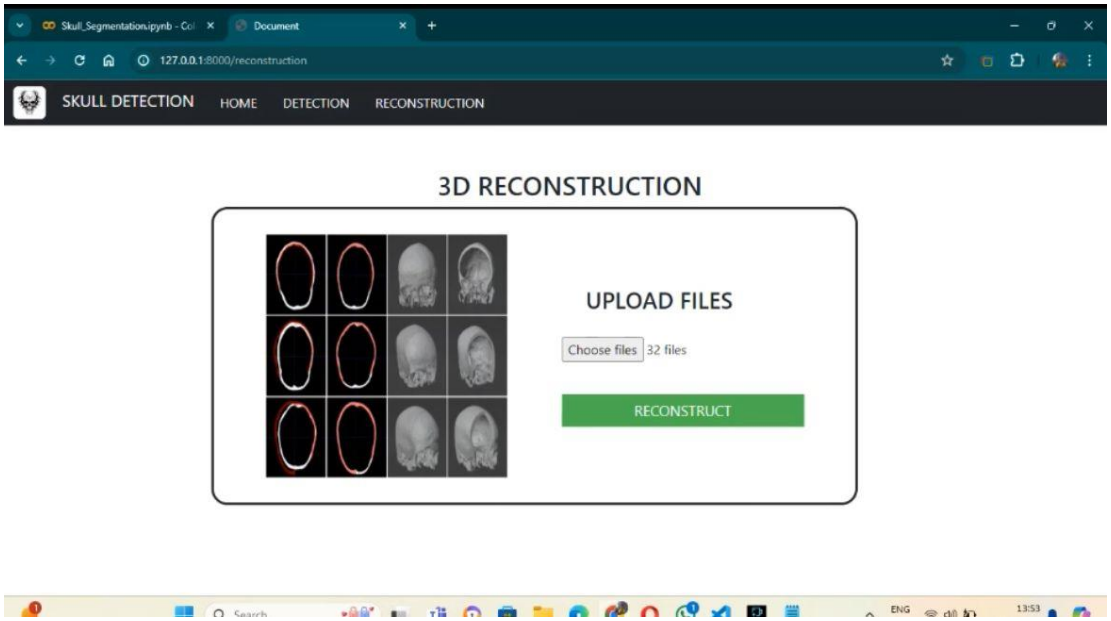

Figure 6.4.4 Skull 3-D Reconstruction Section in Application

The fig 6.4.4 shows the 3D reconstruction section of the application. It has a choose file button where Dicom images of skull should be inserted.

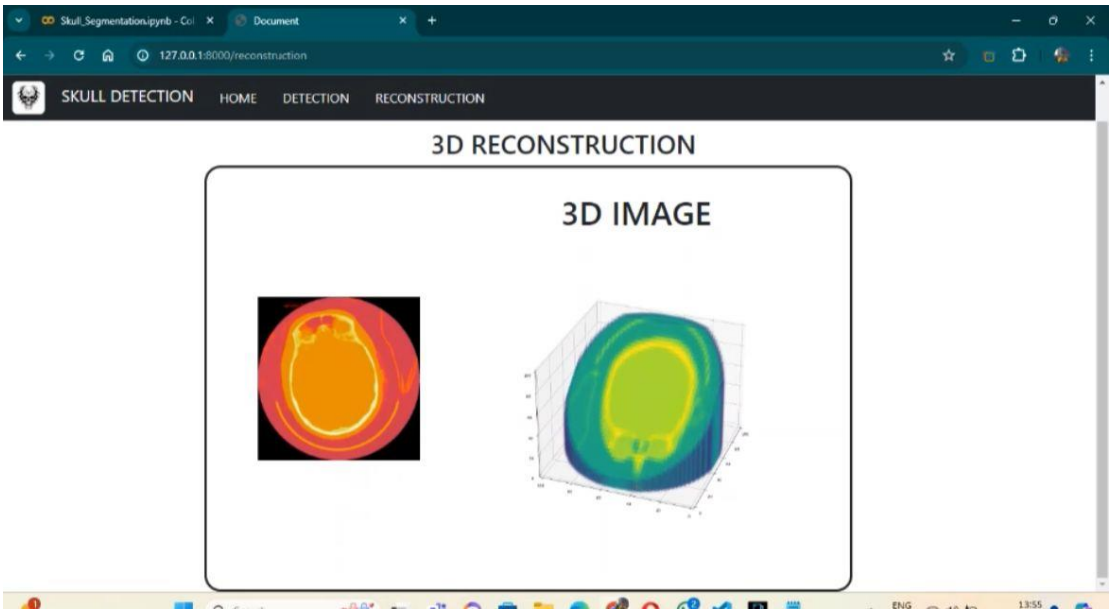

Figure 6.4.5 Skull Fracture Detected in 3-Dimension

The figure 6.4.5 shows the reconstructed skull fracture in 3 dimensional plane with the animated version of the same.

6.5 Comparison with existing work

Detection Accuracy

Our Method: Utilizes YOLOv3, a state-of-the-art object detection algorithm, to achieve high accuracy in skull fracture detection (over 95%).

Existing Methods: Often rely on traditional machine learning algorithms or older neural network architectures, resulting in lower detection accuracy (typically below 95%).

Processing Time

Our Method: YOLOv3 is optimized for speed, enabling faster processing and real-time detection capabilities.

Existing Methods: Traditional methods are slower due to the need for extensive preprocessing and less efficient detection algorithms.

Codebase Size and Complexity

Our Method: The code is concise and leverages modern libraries and techniques, resulting in a smaller and more maintainable codebase.

Existing Methods: Tend to have larger, more complex codebases due to the use of older algorithms and the need for more extensive preprocessing and feature extraction.

Visualization and Reconstruction

Our Method: Provides advanced 3D visualization and reconstruction capabilities with a compact implementation.

Existing Methods: Often involve more manual steps and complex code for achieving 3D visualization, making the process more cumbersome.

Model Training and Deployment

Our Method: YOLOv3 can be trained relatively quickly and deployed efficiently, benefiting from extensive community support and pre-trained weights.

Existing Methods: Training traditional models can be more time-consuming and complex, with fewer pre-trained options available.

Conclusion

Your approach using YOLOv3 for skull fracture detection and 3D reconstruction presents several advantages over existing methods:

Higher Accuracy: Achieving over 95% accuracy.

Faster Processing: Efficient real-time detection.

Compact Code: Easier to maintain and understand.

Advanced Visualization: Effective 3D rendering and segmentation

CONCLUSION

The well-trained DLS system exhibited an excellent diagnostic capability in distinguishing skull fractures under limited conditions. It could be a trusted tool for the detection of skull fractures. More image data in different clinical conditions will be needed in future studies to improve CNN performance. Guiding by the morphological features of the skull, a skeleton-based region proposal method is proposed to make candidate boxes more concentrated in key regions and reduced invalid boxes. With this advantage, the region proposal network in Faster R-CNN is removed for less computation. On the other hand, a novel full resolution feature network is constructed to obtain more precise features to make the model more snesetive to small objects. Experiment results showed that most of skull fractures could be detected correctly by the proposed method in a short time. Compared to the previous works on the skull fracture detection, Skull R-CNN significantly reduces the false positives, and keeps a high sensitivity.

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